Chapter 11

Autoregressive Models and Forecasting

A fading memory, and forecasts as probability distributions

In this chapter. You will meet the autoregressive model of order one, AR(1): the simplest model in which today's value depends on yesterday's. You will learn when it is stationary, how to find its mean, variance, and autocorrelation function, and how to use it to forecast, including how uncertain a forecast is. The chapter ends by placing AR(1) among the other models of the course.

You will use. White noise, stationarity, the ACF, and the random walk (Chapter 10); the Markov property (Session 8); and the rules for expectation, variance, and covariance of sums (Session 5).

How to use these notes. Read them before the exercises. Follow each worked example with pen and paper, and try the *Check* questions before looking at the answers at the end of the chapter.

Contents

11.1 A memory that fades . . . 3
11.2 The AR(1) model . . . 3
11.3 What the coefficient ϕ does . . . 4
11.4 AR(1) is a Markov process . . . 5
11.5 Unrolling the recursion . . . 6
11.5.1 When is AR(1) stationary? . . . 7
11.6 Mean, variance, and ACF . . . 7
11.7 Recognising AR and MA from the ACF . . . 9
11.8 Forecasting . . . 10
11.8.1 The point forecast, step by step . . . 10
11.8.2 The point forecast in one formula . . . 10
11.8.3 How uncertain is the forecast? . . . 11
11.8.4 Checking the formulas by simulation . . . 12
11.9 Long horizons: mean reversion versus the random walk . . . 13
11.10 Updating a forecast . . . 14
11.11 What a forecast interval leaves out . . . 14
11.12 The models of the course side by side . . . 15
11.13 Summary . . . 16
Answers to the checks . . . 16

11.1 A memory that fades

Chapter 10 gave us two extremes. A moving average forgets the past after a fixed number of steps. A random walk never forgets: every shock stays in the series forever. Many real series behave in between.

Think of the hourly CPU load of a server. If the load is high now, it will probably still be high in an hour, because the jobs that cause it are still running. In five hours the current value tells us less, and tomorrow it tells us almost nothing: the load will be back near its usual level. We want a model in which the influence of the present fades gradually.

The idea is to let the next value be a fraction of the current value plus a new, unpredictable shock. Because the series is regressed on its own past, the model is called **autoregressive**.

11.2 The AR(1) model

Definition (AR(1)). Let $\{\varepsilon_t\}$ be white noise with variance σ^2. The process $\{X_t\}$ is an *autoregressive process of order 1* if

$$X_t = c + \phi X_{t-1} + \varepsilon_t, \tag{11.2.1}$$

where c and ϕ are constants and each shock ε_t is uncorrelated with all earlier values $X_{t-1}, X_{t-2}, \dots$

Read the equation from left to right: *the value now* equals *a constant*, plus *a fraction ϕ of the previous value*, plus *something new*. The shock ε_t is often called the **innovation**: it is the part of X_t that could not be predicted from the past.

Example 11.2.1 Running the model by hand

The CPU load (in %) is modelled by $X_t = 12 + 0.6X_{t-1} + \varepsilon_t$. The load starts at $x_0 = 30$, and suppose the shocks happen to be $2, -4, 1, 3, -1$. Each new value uses the previous one:

t	calculation	shock ε_t	x_t
1	$12 + 0.6 \cdot 30$	$+2$	32.00
2	$12 + 0.6 \cdot 32$	-4	27.20
3	$12 + 0.6 \cdot 27.2$	$+1$	29.32
4	$12 + 0.6 \cdot 29.32$	$+3$	32.59
5	$12 + 0.6 \cdot 32.592$	-1	30.56

Without shocks, a load of 30 would stay at 30, since $12 + 0.6 \cdot 30 = 30$. The shocks push the series away from 30, and the factor 0.6 pulls it back. This push and pull is how every AR(1) path is produced.

Check 1. In the same model, the load is $x_0 = 40$ and the next shock is $\varepsilon_1 = -2$. Find x_1.

More lags. The *autoregressive process of order* p, AR(p), uses the p most recent values: $X_t = c + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \varepsilon_t$. In this course we work with AR(1); the ideas carry over.

11.3 What the coefficient ϕ does

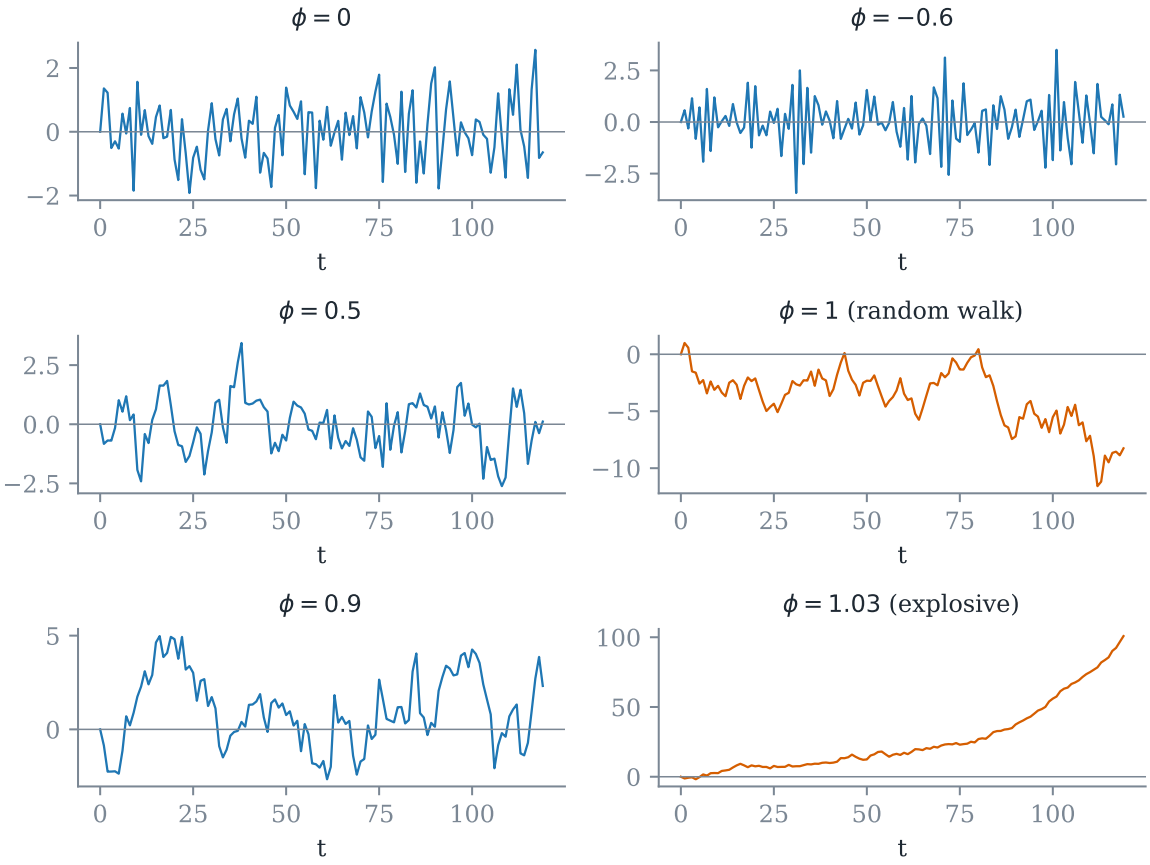


Figure 11.3.1. Simulated AR(1) paths with $c = 0$ and $\sigma = 1$ for six values of ϕ. All use independent shocks.

Figure 11.3.1 shows how much ϕ changes the behaviour.

$\phi = 0$**:** $X_t = c + \varepsilon_t$ is white noise around c. No memory.

$0 < \phi < 1$**:** the series persists: high values follow high values, and it returns to its level more slowly the closer ϕ is to 1. Compare $\phi = 0.5$ with $\phi = 0.9$.

$-1 < \phi < 0$**:** the series tends to alternate above and below its level, producing a zigzag.

$\phi = 1$**:** $X_t = c + X_{t-1} + \varepsilon_t$ is the random walk of Chapter 10 (with drift if $c \neq 0$). There is no level to return to.

$|\phi| > 1$**:** each value amplifies the previous one; the series explodes. Even $\phi = 1.03$ eventually runs away.

Example 11.3.1 The zigzag of a negative ϕ

Let $X_t = -0.5X_{t-1} + \varepsilon_t$ and start at $x_0 = 4$. If no shocks occurred, the values would be

$$4,\ -2,\ 1,\ -0.5,\ 0.25,\ \ldots$$

Each step flips the sign and halves the distance to 0. With shocks added, the path still tends to jump from one side of 0 to the other, as in the $\phi = -0.6$ panel of Figure 11.3.1.

11.4 AR(1) is a Markov process

The future of an AR(1) process depends on the past only through the current value: once we know $X_{t-1} = x$, the next value is $c + \phi x$ plus a new shock, whatever happened earlier. This is the Markov property of Session 8, now with a continuous state space.

If the shocks are normal, $\varepsilon_t \sim N(0, \sigma^2)$, the "transition rule" is a normal distribution:

$$X_t \mid X_{t-1} = x \ \sim\ N\big(c + \phi x,\ \sigma^2\big)\,. \tag{11.4.1}$$

It plays the role of the transition matrix of a Markov chain.

Example 11.4.1 The next hour, given the current load

In the CPU model, $\sigma = 3$. The current load is unusually high: $x_t = 45$. Then

$$X_{t+1} \mid X_t = 45 \ \sim\ N(12 + 0.6 \cdot 45,\ 3^2) = N(39,\ 9),$$

so

$$\mathbb{P}(X_{t+1} > 40 \mid X_t = 45) = \mathbb{P}\left(Z > \frac{40 - 39}{3}\right) = \mathbb{P}(Z > 0.33) \approx 0.37.$$

Without the information $x_t = 45$, the load in a random hour has the stationary distribution $N(30, 3.75^2)$ (derived in Section 11.6), and $\mathbb{P}(X > 40) \approx 0.004$. Knowing the current value changes the answer by a factor of almost 100 (Figure 11.4.1).

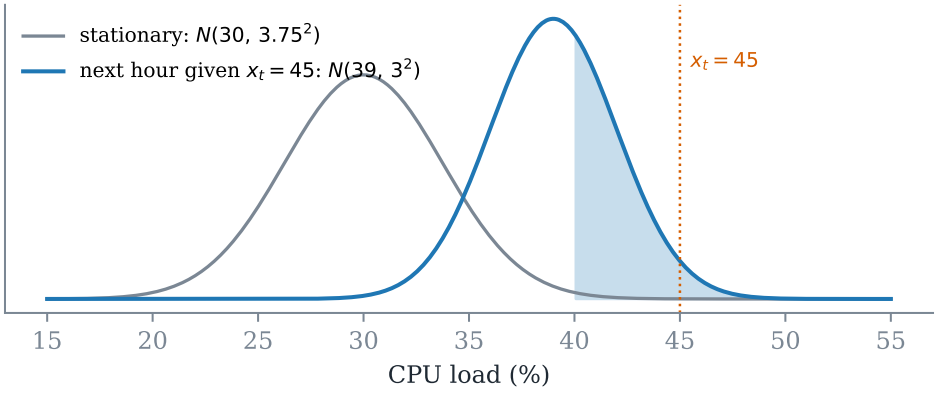


Figure 11.4.1. The distribution of next hour's load given $x_t = 45$, compared with the stationary distribution. The shaded area is $\mathbb{P}(X_{t+1} > 40 \mid X_t = 45)$.

A simulation of AR(1) follows the definition step by step:

```
import numpy as np

def simulate_ar1(c, phi, sigma, n, x0, rng):
    x = np.empty(n)
    x[0] = x0
    for t in range(1, n):
        x[t] = c + phi * x[t-1] + rng.normal(0, sigma)
    return x

rng = np.random.default_rng(11)
path = simulate_ar1(c=12, phi=0.6, sigma=3, n=500, x0=30, rng=rng)
```

11.5 Unrolling the recursion

Where does the memory of an AR(1) process come from? Substitute the equation into itself. One step back:

$$X_t = c + \phi\big(c + \phi X_{t-2} + \varepsilon_{t-1}\big) + \varepsilon_t = c(1+\phi) + \phi^2 X_{t-2} + \varepsilon_t + \phi\,\varepsilon_{t-1}.$$

Repeating this k times gives

$$X_t = c\left(1 + \phi + \cdots + \phi^{k-1}\right) + \phi^k X_{t-k} + \varepsilon_t + \phi\,\varepsilon_{t-1} + \phi^2\varepsilon_{t-2} + \cdots + \phi^{k-1}\varepsilon_{t-k+1}. \tag{11.5.1}$$

A shock that happened j steps ago is still present today, with weight ϕ^j. Figure 11.5.1 shows these weights.

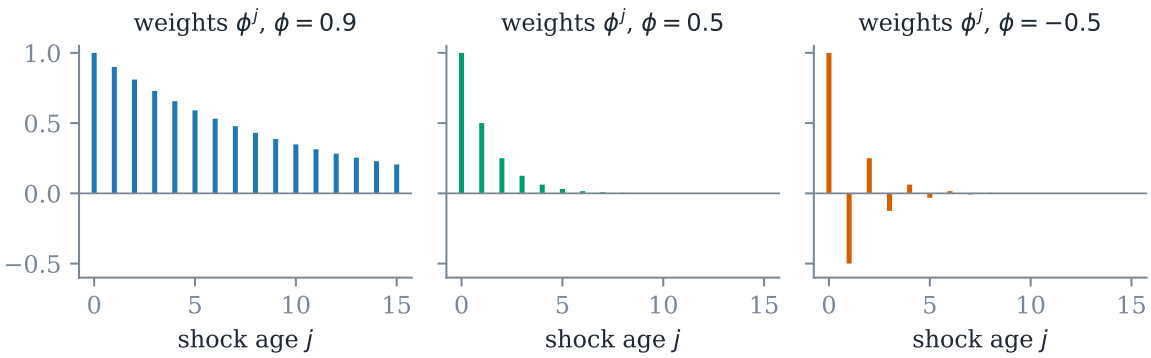


Figure 11.5.1. The weight ϕ^j with which a shock from j steps ago enters today's value.

Example 11.5.1 How long does a shock last?

In the CPU model ($\phi = 0.6$), a sudden extra load of 10 percentage points contributes $10 \cdot 0.6 = 6$ points one hour later, $10 \cdot 0.6^2 = 3.6$ two hours later, and only $10 \cdot 0.6^5 \approx 0.78$ after five hours.

With $\phi = 0.9$ the same shock still contributes $10 \cdot 0.9^5 \approx 5.9$ points after five hours. A larger ϕ means a longer memory.

11.5.1 When is AR(1) stationary?

Equation (11.5.1) answers this question.

- If $|\phi| < 1$, the weights ϕ^j shrink towards zero. The starting value is forgotten ($\phi^k X_{t-k} \to 0$), the constants add up to $c/(1-\phi)$, and in the limit
$$X_t = \frac{c}{1-\phi} + \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}. \tag{11.5.2}$$
This is a weighted sum of white noise with weights that do not depend on t, so the process is stationary.
- If $\phi = 1$, every old shock keeps weight 1. This is the random walk, whose variance grows like t.
- If $|\phi| > 1$, old shocks get larger weights than new ones, and the process explodes.

Result (stationarity of AR(1)). The AR(1) process is stationary if and only if $-1 < \phi < 1$.

Key idea. Equation (11.5.2) shows that a stationary AR(1) is a moving average of white noise with *infinitely many* terms whose weights fade geometrically. A moving average forgets abruptly; an AR(1) process forgets gradually.

Check 2. Which of the following give a stationary AR(1) process: $\phi = 0.99$, $\phi = -1$, $\phi = 1.2$, $\phi = -0.3$?

11.6 Mean, variance, and ACF

Assume from now on that $|\phi| < 1$. Because the process is stationary, $\mathbb{E}[X_t]$ and $\mathrm{Var}(X_t)$ are the same at all times. This gives a very short way to find them: apply $\mathbb{E}$ or Var to both sides of the model.

The mean

Take expectations of $X_t = c + \phi X_{t-1} + \varepsilon_t$. With $\mathbb{E}[X_t] = \mathbb{E}[X_{t-1}] = \mu$ and $\mathbb{E}[\varepsilon_t] = 0$,
$$\mu = c + \phi\mu \qquad \Longrightarrow \qquad \mu = \frac{c}{1-\phi}.$$
Subtracting $\mu = c + \phi\mu$ from the model gives the **centred form**
$$X_t - \mu = \phi\,(X_{t-1} - \mu) + \varepsilon_t. \tag{11.6.1}$$
In words: today's deviation from the mean is the fraction ϕ of yesterday's deviation, plus a shock. This form is the easiest to use for calculations.

The variance

ε_t is uncorrelated with X_{t-1}, so the variance of the sum is the sum of the variances. With $\mathrm{Var}(X_t) = \mathrm{Var}(X_{t-1}) = \gamma(0)$,
$$\gamma(0) = \phi^2\gamma(0) + \sigma^2 \qquad \Longrightarrow \qquad \gamma(0) = \frac{\sigma^2}{1-\phi^2}.$$

The autocovariance and ACF

For $h \geq 1$, use the centred form at time $t+h$ and take the covariance with X_t. The shock ε_{t+h} happens after time t, so it is uncorrelated with X_t:

$$\gamma(h) = \operatorname{Cov}(X_{t+h}, X_t) = \phi \operatorname{Cov}(X_{t+h-1}, X_t) + 0 = \phi\,\gamma(h-1).$$

Each extra lag multiplies the covariance by ϕ, so $\gamma(h) = \phi^h \gamma(0)$.

Result (moments of a stationary AR(1)). For $|\phi| < 1$,

$$\mu = \frac{c}{1-\phi}, \qquad \gamma(0) = \frac{\sigma^2}{1-\phi^2}, \qquad \rho(h) = \phi^{|h|}. \tag{11.6.2}$$

Example 11.6.1 The CPU model

For $X_t = 12 + 0.6X_{t-1} + \varepsilon_t$ with $\sigma = 3$:

$$\mu = \frac{12}{0.4} = 30, \qquad \gamma(0) = \frac{9}{1-0.36} = 14.0625, \qquad \sqrt{\gamma(0)} = 3.75,$$

and $\rho(1) = 0.6$, $\rho(2) = 0.36$, $\rho(3) = 0.216$. A simulation of 200 000 hours gave mean 29.98, variance 13.96, and sample autocorrelations 0.60, 0.36, 0.22.

Example 11.6.2 A strongly persistent series

Let $X_t = 4 + 0.8X_{t-1} + \varepsilon_t$ with $\sigma^2 = 1$. Then

$$\mu = \frac{4}{0.2} = 20, \qquad \gamma(0) = \frac{1}{1-0.64} = 2.78, \qquad \rho(3) = 0.8^3 = 0.512.$$

Values three steps apart still have correlation above one half.

Example 11.6.3 A negative coefficient

Let $X_t = 6 - 0.5X_{t-1} + \varepsilon_t$ with $\sigma^2 = 3$. Then

$$\mu = \frac{6}{1-(-0.5)} = 4, \qquad \gamma(0) = \frac{3}{1-0.25} = 4, \qquad \rho(1) = -0.5, \quad \rho(2) = 0.25, \quad \rho(3) = -0.125.$$

Note the denominator $1 - \phi = 1.5$: with a negative ϕ, the mean is *smaller* than c.

Key idea. The series varies *more* than the shocks: $\gamma(0) > \sigma^2$ unless $\phi = 0$. Dependence lets shocks accumulate before they fade. As ϕ approaches ± 1, the variance grows without bound (Figure 11.6.1), on the way to the random walk.

Watch out. Two common mix-ups. (1) c is not the mean unless $\phi = 0$; the mean is $c/(1-\phi)$. (2) σ^2 is the variance of the *shocks*, not of the series; the series has variance $\sigma^2/(1-\phi^2)$.

Check 3. For $X_t = 10 + 0.5X_{t-1} + \varepsilon_t$ with $\sigma^2 = 3$, find μ, $\gamma(0)$, and $\rho(2)$.

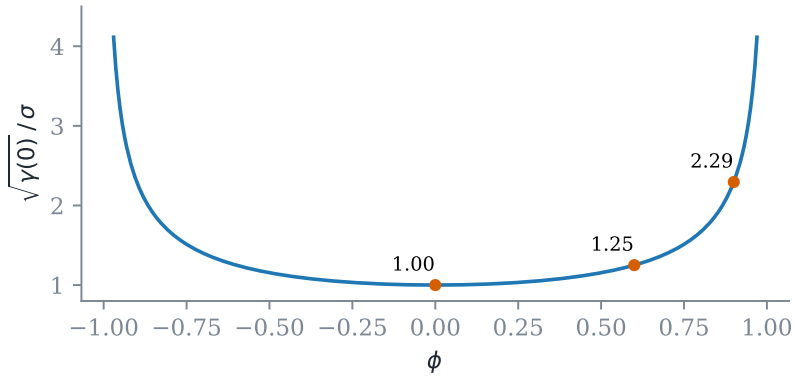


Figure 11.6.1. The standard deviation of a stationary AR(1), relative to the shock standard deviation σ, as a function of ϕ.

11.7 Recognising AR and MA from the ACF

The ACF of an AR(1) *decays* geometrically but never becomes exactly zero. The ACF of a moving average *cuts off*: it is exactly zero beyond a certain lag. Figure 11.7.1 shows the contrast.

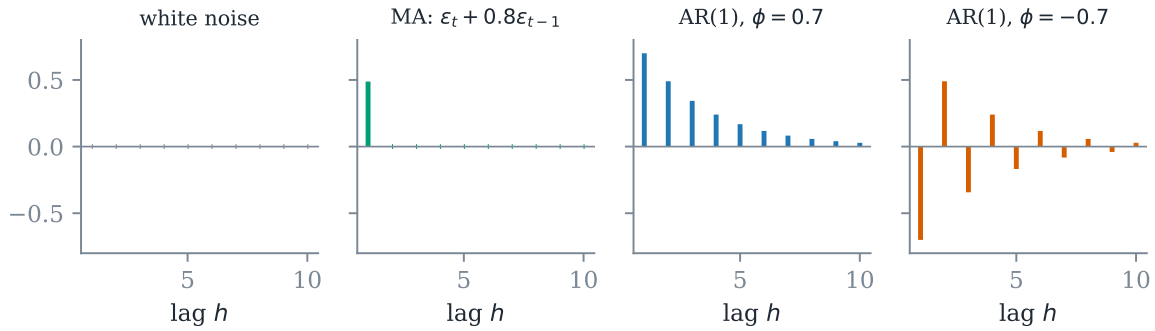


Figure 11.7.1. Theoretical ACFs of four stationary models. For the moving average, $\rho(1) = 0.8/(1 + 0.8^2) \approx 0.49$.

Model	**ACF $\rho(h)$, $h \geq 1$**
White noise	0 at all lags
Moving average over $q + 1$ terms	non-zero up to lag q, then exactly 0
AR(1), $0 < \phi < 1$	ϕ^h: positive, decaying geometrically
AR(1), $-1 < \phi < 0$	ϕ^h: alternating signs, decaying

Example 11.7.1 Which model fits this ACF?

A stationary series has sample autocorrelations $r_1 = 0.62$, $r_2 = 0.41$, $r_3 = 0.24$, $r_4 = 0.17$, $r_5 = 0.08$. They decrease gradually instead of dropping to zero after a few lags, and most of them are roughly 0.6 to 0.7 times the previous one. This is the pattern of an

AR(1) with ϕ around 0.6 to 0.65, not of a short moving average. Remember from Chapter 10 that each r_k also contains sampling variation.

11.8 Forecasting

We have observed the series up to time T and want to say something about X_{T+h}, the value h steps ahead. The number h is the **forecast horizon**.

Key idea. A forecast is not one number. It is the *conditional distribution* of the future value given what has been observed. We summarise it by its mean (the point forecast) and an interval that contains the future value with a stated probability.

Because AR(1) is a Markov process, only the last observation x_T matters. We write $\hat{x}_{T+h|T}$ for the forecast of X_{T+h} made at time T.

11.8.1 The point forecast, step by step

The simplest method follows the model forward, replacing each unknown future shock by its mean, 0:

$$\begin{aligned}\hat{x}_{T+1|T} &= c + \phi\, x_T,\\ \hat{x}_{T+2|T} &= c + \phi\, \hat{x}_{T+1|T},\\ \hat{x}_{T+3|T} &= c + \phi\, \hat{x}_{T+2|T}, \quad \text{and so on.}\end{aligned}$$

Example 11.8.1 Forecasting the CPU load, step by step

In the CPU model, the last observation is $x_T = 45$.

$$\begin{aligned}\hat{x}_{T+1|T} &= 12 + 0.6 \cdot 45 = 39,\\ \hat{x}_{T+2|T} &= 12 + 0.6 \cdot 39 = 35.4,\\ \hat{x}_{T+3|T} &= 12 + 0.6 \cdot 35.4 = 33.24.\end{aligned}$$

The forecasts move from 45 back towards the mean 30.

11.8.2 The point forecast in one formula

Using the centred form h times,

$$X_{T+h} - \mu = \phi^h (x_T - \mu) + \underbrace{\varepsilon_{T+h} + \phi\, \varepsilon_{T+h-1} + \cdots + \phi^{h-1} \varepsilon_{T+1}}_{\text{future shocks}}. \tag{11.8.1}$$

The first term is known at time T; the future shocks have mean 0. Therefore

$$\hat{x}_{T+h|T} = \mathbb{E}[X_{T+h} \mid X_T = x_T] = \mu + \phi^h\, (x_T - \mu). \tag{11.8.2}$$

For the CPU example, $\hat{x}_{T+h|T} = 30 + 0.6^h \cdot 15$, which reproduces 39, 35.4, and 33.24.

11.8.3 How uncertain is the forecast?

The future shocks in (11.8.1) are uncorrelated, so their variances add:

$$\sigma_h^2 = \operatorname{Var}(X_{T+h} \mid X_T = x_T) = \sigma^2 \left(1 + \phi^2 + \phi^4 + \cdots + \phi^{2(h-1)}\right) = \sigma^2 \frac{1-\phi^{2h}}{1-\phi^2}. \tag{11.8.3}$$

One step ahead, $\sigma_1^2 = \sigma^2$: only one new shock is unknown. Every further step adds another shock, but with a smaller weight.

Result (AR(1) forecast distribution). If the shocks are $N(0, \sigma^2)$, then given $X_T = x_T$,

$$X_{T+h} \;\sim\; N\!\left(\mu + \phi^h (x_T - \mu),\;\; \sigma^2 \frac{1-\phi^{2h}}{1-\phi^2}\right),$$

and a 95% forecast interval is $\hat{x}_{T+h|T} \pm 1.96\,\sigma_h$.

Example 11.8.2 Forecast intervals for the CPU load

Continuing Example 11.8.1 with $\sigma = 3$:

h	$\hat{x}_{T+h\|T}$	σ_h^2	σ_h	95% interval
1	39.00	9	3.00	[33.1, 44.9]
2	35.40	$9(1+0.36) = 12.24$	3.50	[28.5, 42.3]
3	33.24	$9(1+0.36+0.1296) = 13.41$	3.66	[26.1, 40.4]
5	31.17	13.98	3.74	[23.8, 38.5]
∞	30.00	14.06	3.75	[22.65, 37.35]

After a few hours the forecast is essentially the stationary distribution $N(30, 3.75^2)$.

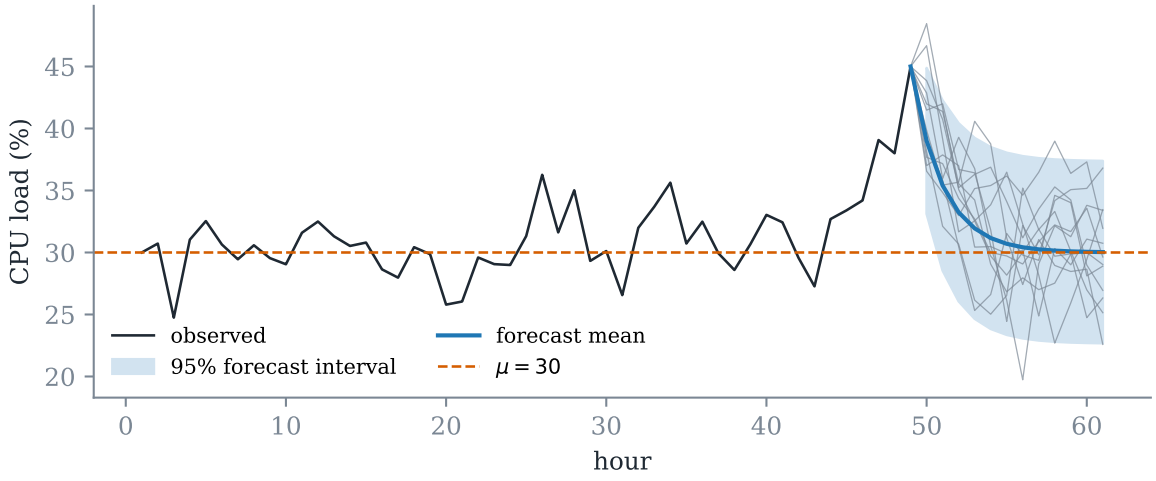


Figure 11.8.1. The CPU forecast from $x_T = 45$: forecast mean, 95% interval, and 12 of 5000 simulated futures (grey).

11.8.4 Checking the formulas by simulation

We can test the result by simulating many futures from $x_T = 45$ and reading off the empirical 2.5% and 97.5% quantiles at each horizon.

```
rng = np.random.default_rng(2026)
c, phi, sigma, xT, H, reps = 12, 0.6, 3, 45, 10, 5000
x = np.full(reps, float(xT))
sims = np.empty((reps, H))
for h in range(H):
    x = c + phi * x + rng.normal(0, sigma, reps)
    sims[:, h] = x
lo, hi = np.quantile(sims, [0.025, 0.975], axis=0)
```

Horizon	1	2	5	10
Theory	[33.1, 44.9]	[28.5, 42.3]	[23.8, 38.5]	[22.7, 37.4]
Simulation	[33.1, 44.8]	[28.5, 42.0]	[24.1, 38.4]	[22.7, 37.2]

The agreement is within simulation error. The simulation also works for models where no formula is available.

Example 11.8.3 Forecasting with a negative coefficient

Let $\mu = 0$, $\phi = -0.5$, $\sigma = 1$, and $x_T = 4$. The point forecasts $0 + (-0.5)^h \cdot 4$ are

$$-2,\ 1,\ -0.5,\ 0.25,\ \ldots,$$

alternating around the mean while shrinking (Figure 11.8.2). The forecast standard deviation grows from $\sigma_1 = 1$ to $1/\sqrt{1 - 0.25} \approx 1.155$.

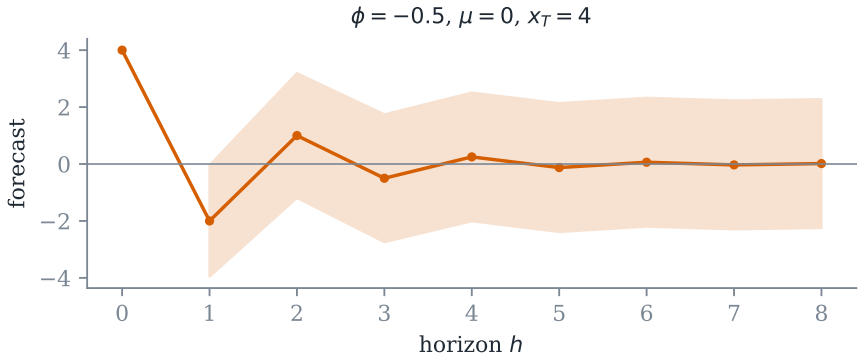


Figure 11.8.2. Forecasts and 95% intervals for an AR(1) with $\phi = -0.5$.

Check 4. For the model of Check 3, suppose $x_T = 26$. Find the one-step forecast with its 95% interval, and the two-step forecast with its variance.

11.9 Long horizons: mean reversion versus the random walk

For a stationary AR(1), as h grows,

$$\hat{x}_{T+h|T} \to \mu \qquad \text{and} \qquad \sigma_h^2 \to \frac{\sigma^2}{1-\phi^2} = \gamma(0).$$

The forecast forgets the last observation and becomes the stationary distribution. The expected deviation from the mean, $\phi^h(x_T - \mu)$, shrinks by the factor ϕ at every step. This is called **mean reversion**.

Example 11.9.1 How quickly is the last observation forgotten?

For $\phi = 0.6$, the expected deviation after h steps is the fraction 0.6^h of today's deviation: $0.6^4 \approx 0.13$ and $0.6^5 \approx 0.08$. So after 5 hours less than 10% of today's deviation remains. For $\phi = 0.9$: $0.9^{21} \approx 0.11$ and $0.9^{22} \approx 0.098$, so it takes 22 steps to get below 10%.

For the **random walk** ($\phi = 1$) the forecast formulas change:

$$\hat{x}_{T+h|T} = x_T, \qquad \sigma_h^2 = h\sigma^2.$$

The best forecast is the last observed value, and the uncertainty grows without limit.

Example 11.9.2 Forecasting a backlog

A backlog behaves like a random walk with $\sigma = 2$ tickets per day and today has $x_T = 120$ open tickets. The forecast for 25 days ahead is 120, with $\sigma_{25} = 2\sqrt{25} = 10$ and 95% interval 120 ± 19.6, that is $[100.4,\ 139.6]$. In 100 days the interval is twice as wide.

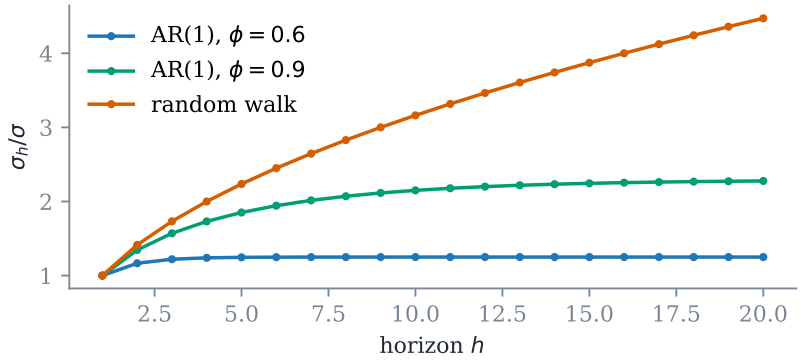


Figure 11.9.1. Forecast standard deviation relative to σ: it levels off for a stationary AR(1) and keeps growing for a random walk.

Key idea. Whether a series is modelled as stationary or as a random walk hardly matters for a one-step forecast, but it matters a great deal for long-range forecasts.

Check 5. A random walk has $\sigma = 3$ and $x_T = 50$. Give the 95% forecast interval four steps ahead.

11.10 Updating a forecast

When a new value is observed, there is no need to adjust the old forecast by hand. Because the model is Markov, we simply start the forecasting recursion again from the newest observation.

Example 11.10.1 The CPU load one hour later

At hour T the forecast for hour $T+1$ was 39. The observed value turns out to be $x_{T+1} = 34$. The difference $34 - 39 = -5$ is the shock ε_{T+1} that actually happened.
Starting again from 34, the new forecasts are

$$\hat{x}_{T+2|T+1} = 30 + 0.6 \cdot (34 - 30) = 32.4, \qquad \hat{x}_{T+3|T+1} = 30 + 0.6^2 \cdot 4 = 31.44,$$

compared with the old forecasts 35.4 and 33.24. The new forecast for $T+2$ is a one-step forecast, so its variance is back to $\sigma^2 = 9$ instead of 12.24.
The old and new forecasts for $T+2$ differ by $32.4 - 35.4 = -3 = 0.6 \cdot (-5)$: the surprise at $T+1$ is passed on with the usual weight ϕ.

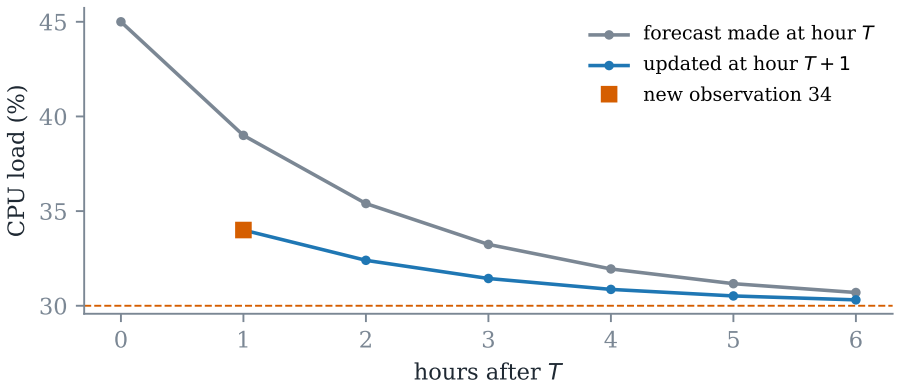


Figure 11.10.1. Forecasts made at hour T (grey) and updated at hour $T+1$ after observing 34 (blue).

11.11 What a forecast interval leaves out

The intervals above are exact *if the model is true and its parameters are known*. In practice three sources of uncertainty are missing.

Parameter uncertainty. In applications c, ϕ, and σ are estimated from data, and the estimates are uncertain. Estimation belongs to statistics and is not part of this course.

Model uncertainty. The data might be better described by another model: a random walk instead of an AR(1), or a model with a trend or seasonality.

Changes in the system. The normal intervals assume uncorrelated normal shocks with constant variance and a process that does not change. A new release, a marketing campaign, or a holiday can break these assumptions.

For these reasons, model-based forecast intervals tend to be too narrow in practice.

Key idea. A useful first check of a model is to simulate several paths from it and compare them with the observed series. Do they have the same level, the same spread, and the same persistence? Does the model ACF $\phi^{|h|}$ resemble the sample ACF, given the sampling variation of Chapter 10? If the simulated paths look nothing like the data, the model is wrong, however good its formulas are.

11.12 The models of the course side by side

AR(1) is the last model of the course. Each model answers a different type of question.

Model	Time and values	Dependence	Typical question
Binomial	one experiment; a count	independent trials	How many of 200 calls fail?
Poisson process	continuous time; counts	independent increments	When does the next request arrive?
Markov chain	discrete time; a few states	next state depends on current state	What fraction of time is the service down?
Random walk	discrete time; real values	every shock is kept	How far can the backlog drift in a month?
AR(1)	discrete time; real values	influence fades by ϕ per step	What will the load be in five hours?

Example 11.12.1 Choosing a model

Commits to a repository by many independent developers. We count events in continuous time, and commits by different people are roughly independent: a Poisson process, possibly with a rate that changes over the day.
The state of a build server: idle, building, or failed, recorded every hour. A few discrete states, with the next state depending on the current one: a Markov chain.
Daily average memory use that fluctuates around a stable level with persistence. Real values, a fixed level, and fading dependence: an AR(1).
The net change in the number of users, built up from independent daily gains and losses. Every daily change is kept: a random walk.
In each case, ask which assumptions could fail. Are the developers really independent? Does the build server remember more than its current state? Is the memory level really stable over months?

Key idea. Three of these models share one structure: the future depends on the past only through the present. The Poisson process starts afresh after each event, a Markov chain moves according to its current state, and an AR(1) process steps forward from its current value. This Markov idea is what makes the forecasts and simulations of the course possible.

11.13 Summary

Concept	What to remember
AR(1)	$X_t = c + \phi X_{t-1} + \varepsilon_t$
Behaviour of ϕ	0: white noise; $(0,1)$: persistence; $(-1,0)$: zigzag; 1: random walk
Stationarity	$-1 < \phi < 1$
Moments	$\mu = c/(1-\phi)$, $\gamma(0) = \sigma^2/(1-\phi^2)$, $\rho(h) = \phi^{\lvert h\rvert}$
Centred form	$X_t - \mu = \phi(X_{t-1} - \mu) + \varepsilon_t$
Point forecast	$\hat{x}_{T+h\mid T} = \mu + \phi^h(x_T - \mu)$
Forecast variance	$\sigma_h^2 = \sigma^2(1-\phi^{2h})/(1-\phi^2)$; random walk: $h\sigma^2$
95% interval	$\hat{x}_{T+h\mid T} \pm 1.96\,\sigma_h$ for normal shocks
Updating	Restart the recursion from the newest observation

Answers to the checks

Check 1. $x_1 = 12 + 0.6 \cdot 40 - 2 = 34$.

Check 2. $\phi = 0.99$ and $\phi = -0.3$ are stationary. $\phi = -1$ and $\phi = 1.2$ are not, since stationarity requires $-1 < \phi < 1$ strictly.

Check 3. $\mu = 10/(1-0.5) = 20$, $\gamma(0) = 3/(1-0.25) = 4$, and $\rho(2) = 0.5^2 = 0.25$.

Check 4. $\hat{x}_{T+1\mid T} = 10 + 0.5 \cdot 26 = 23$, with $\sigma_1 = \sqrt{3} \approx 1.73$ and 95% interval $23 \pm 3.39 = [19.6,\ 26.4]$. Two steps: $\hat{x}_{T+2\mid T} = 20 + 0.25 \cdot 6 = 21.5$, with variance $3(1 + 0.25) = 3.75$.

Check 5. $\sigma_4 = 3\sqrt{4} = 6$, so $50 \pm 1.96 \cdot 6 = [38.2,\ 61.8]$.